

AWS EBS (Elastic Block Store).

1. What is AWS EBS?

AWS EBS (Elastic Block Store) is a block-level storage service provided by Amazon Web Services that is used with Amazon EC2 instances.

Think of it like a virtual hard disk attached to a cloud server.

- It stores persistent data.
- Even if the EC2 instance stops or crashes, the data remains safe.
- Mainly used for databases, file systems, and applications that require high performance storage.

2. Key Characteristics

1. Block Storage

Data is stored in blocks (similar to SSD/HDD).

2. Persistent Storage

Data remains even after instance restart.

3. Attach/Detach Feature

EBS volumes can be attached or detached from EC2 instances.

4. Snapshot Support

Backup of EBS volumes can be stored in Amazon S3 as snapshots.

5. High Availability

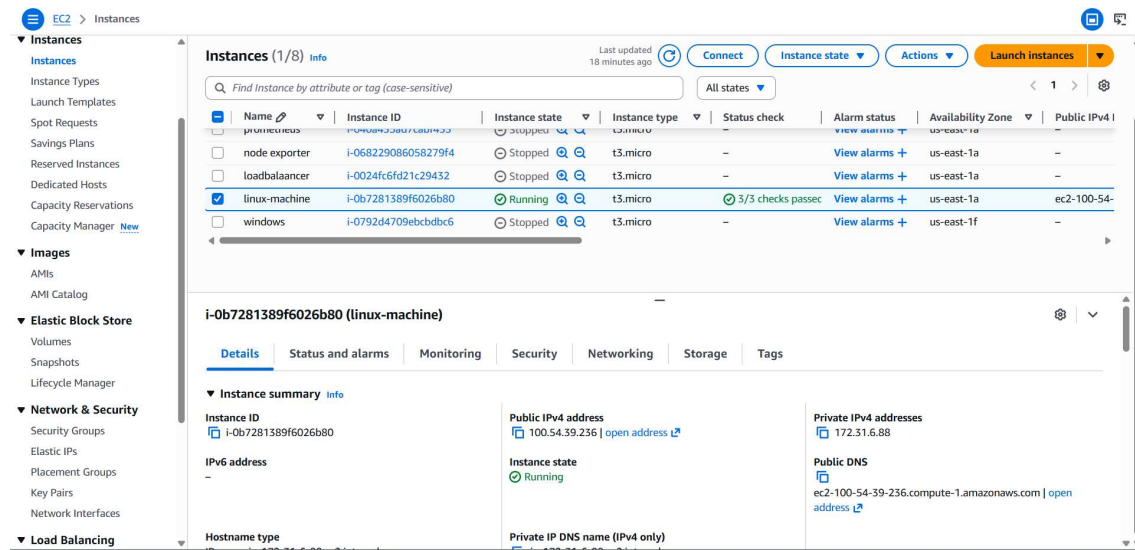
Replicated automatically inside the same Availability Zone.

3. Types of EBS Volumes

Volume Type	Description	Use Case
gp3 / gp2	General Purpose SSD	Web servers, small databases
io1 / io2	Provisioned IOPS SSD	High-performance databases
st1	Throughput Optimized HDD	Big data, analytics

Volume Type	Description	Use Case
sc1	Cold HDD	Infrequent access

I have created an instance and then I have created an volume of 100 GB and attached to this instance.



You can check using the **lsblk** command.

```
(ec2-user@ip-172-31-6-88 ~]$ sudo su
[root@ip-172-31-6-88 ec2-user]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1     259:10   0    8G  0 disk 
└─nvme0n1p1 259:11   0    8G  0 part /
└─nvme0n1p2 259:12   0   1M  0 part 
└─nvme0n1p3 259:13   0   10M  0 part /boot/efi
nvme1n1     259:4    0  100G  0 disk
```

This is the created configuration storage of the new volume.

EC2 > Volumes > Create volume

Volume settings

Volume type [Info](#)
General Purpose SSD (gp3)

Size (GiB) [Info](#)
100
Min: 1 GiB, Max: 65536 GiB.

IOPS [Info](#)
3000
Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) [Info](#)
125
Min: 125 MiB, Max: 2000 MiB, Baseline: 125 MiB/s.

Availability Zone [Info](#)
use1-az1 (us-east-1a)

Snapshot ID - optional [Info](#)
Don't create volume from a snapshot

Encryption [Info](#)
Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.
☐ Encrypt this volume

Created volume.

Images
AMIs
AMI Catalog

Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

Network & Security
Security Groups
Elastic IPs
Placement Groups

vol-066e46bc35b255fcd gp3 100 GiB 3000 125 - - 2026/01

Fault tolerance for all volumes in this Region

Snapshot summary
Recently backed up volumes / Total # volumes
0 / 8

Last updated on Mon, Mar 09, 2026, 11:23:40 PM (GMT+05:30)

Data Lifecycle Manager default policy for EBS Snapshots status
No default policy set up | [Create policy](#)

Then we have to attach to the instance but the **EBS** and **EC2** have to be created in same region and same AZ's. Then we have to select the running ec2 machine.

EC2 > Volumes > vol-066e46bc35b255fcd > Attach volume

Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID
vol-066e46bc35b255fcd

Availability Zone
use1-az1 (us-east-1a)

Instance [Info](#)
i-0b7281389f6026b80 (linux-machine) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)
/dev/sdb

Recommended device names for Linux: /dev/xvda for root volume, /dev/sd[f-p] for data volumes.

ⓘ Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.

[Cancel](#) [Attach volume](#)

When we use **lsblk** we can see the un mounted volume with free space.

```
(ec2-user@ip-172-31-6-88 ~)$ sudo su
[root@ip-172-31-6-88 ec2-user]# lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1      259:10   0    8G  0 disk 
├─nvme0n1p1  259:1   0    8G  0 part /
├─nvme0n1p2  259:2   0    1M  0 part 
└─nvme0n1p3  259:3   0   10M  0 part /boot/efi
nvme1n1      259:4   0 100G  0 disk 
[root@ip-172-31-6-88 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs         4.0M    0  4.0M   0% /dev
tmpfs            459M    0  459M   0% /dev/shm
tmpfs           184M  440K  183M   1% /run
/dev/nvme0n1p1    8.0G  1.6G   6.4G  20% /
tmpfs            459M    0  459M   0% /tmp
/dev/nvme0n1p3   10M   1.3M   8.7M  13% /boot/efi
tmpfs            92M    0   92M   0% /run/user/1000
[root@ip-172-31-6-88 ec2-user]#
```

file -s /dev/nvme1n1 >>>> Checks if the disk already has a filesystem.

mkfs -t xfs /dev/nvme1n1 >>>> Creates a filesystem on the disk.

```
[root@ip-172-31-6-88 ec2-user]# mkdir /mnt/hari
[root@ip-172-31-6-88 ec2-user]# file -s /dev/nvme1n1
bash: file -s: command not found
[root@ip-172-31-6-88 ec2-user]# file -s /dev/nvme1n1
/dev/nvme1n1: data
[root@ip-172-31-6-88 ec2-user]# mkfs -t xfs /dev/nvme1n1
meta-data=/dev/nvme1n1          isize=512    agcount=16, agsize=1638400 blks
                     =                       sectsz=512   attr=2, projid32bit=1
                     =                       crc=1     finobt=1, sparse=1, rmapbt=0
                     =                       reflink=1  bigtime=1 inobtcount=1 nrext64=0
                     =                       exchange=0
data                =                       bsize=4096   blocks=26214400, imaxpct=25
                     =                       sunit=1   swidth=1 blks
naming              =version 2              bsize=4096   ascii-ci=0, ftype=1, parent=0
log                  =internal log          bsize=4096   blocks=16384, version=2
                     =                       sectsz=512   sunit=1 blks, lazy-count=1
realtime            =none                   extsz=4096   blocks=0, rtextents=0
[root@ip-172-31-6-88 ec2-user]#
```

i-0b7281389f6026b80 (linux-machine)
PublicIPs: 100.54.39.236 PrivateIPs: 172.31.6.88

mkdir /mnt/myvolume Creates a **mount point directory** where the volume will be attached.

mount /dev/nvme1n1 /mnt/myvolume Mounts the EBS volume to the directory. **df -h** Shows disk usage and mounted filesystems.

```
[root@ip-172-31-6-88 ec2-user]# mount /dev/nvme1n1 /mnt/hari
[root@ip-172-31-6-88 ec2-user]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs         4.0M    0  4.0M   0% /dev
tmpfs            459M    0  459M   0% /dev/shm
tmpfs           184M  440K  183M   1% /run
/dev/nvme0n1p1    8.0G  1.6G   6.4G  20% /
tmpfs            459M    0  459M   0% /tmp
/dev/nvme0n1p3   10M   1.3M   8.7M  13% /boot/efi
tmpfs            92M    0   92M   0% /run/user/1000
/dev/nvme1n1     100G  747M  100G   1% /mnt/hari
[root@ip-172-31-6-88 ec2-user]#
```

i-0b7281389f6026b80 (linux-machine)
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